

AZURE DATA ENGINEERING PROJECT

End-to-End Sales Analytics Pipeline

Building a Centralized Data Warehouse with Azure Data Factory & Medallion Architecture



Azure Data Factory



Azure Data Lake
Storage Gen2



Azure SQL
Database



Power BI



KEY OUTCOMES



Automated
Data Pipeline



Medallion
Architecture



Business-Ready
Dataset



Analytics-Driven
Insights

1. Executive Summary

This project presents the development of an end-to-end sales data pipeline on Microsoft Azure, designed to integrate, transform, and deliver data for analytics and reporting. The solution uses Azure Data Factory for orchestration, Azure Data Lake Storage Gen2 for layered data storage following the Medallion Architecture (Bronze, Silver, Gold), and Azure SQL Database as the serving layer.

The pipeline automates data ingestion from multiple sources, applies data cleansing and transformation, and produces a consolidated, analytics-ready dataset. The final output includes key business metrics such as TotalRevenue, TotalCost, and GrossProfit, enabling efficient reporting in tools like Power BI.

2. Project Objectives

The primary objectives of this project are:

- 1. Build a centralized, End-to-End Sales Analytics Pipeline (Azure Data Warehouse + ADF)
- 2. Integrate and harmonize sales data from multiple heterogeneous sources
- 3. Implement the Medallion Architecture to ensure structured data processing across layers
- 4. Develop reusable and parameterized ADF pipelines for automated data ingestion and transformation
- 5. Generate a Gold-layer dataset optimized for reporting in Power BI and other BI tools
- 6. Apply best practices in Azure Data Engineering, including linked services, integration runtimes, data flows, and pipeline orchestration

3. Solution Architecture

The solution follows the Medallion Architecture pattern — a widely adopted layered approach in modern data engineering. Data flows from source systems through three progressive quality tiers:

Layer	Storage Location	Description
Raw / Source	Blob Storage (denisstorageblob) & ADLS	Original, unmodified source files from all systems
Bronze	ADLS: denis/medallion/bronze/	Ingested files copied as-is with standardized naming
Silver	ADLS: denis/medallion/silver/	Cleaned, validated, and conformed data (one file per entity)
Gold	ADLS: denis/medallion/gold/	Business-ready denormalized dataset with calculated KPIs

3.1 Architecture Diagram

The high-level architecture spans four conceptual layers: Source Layer, Ingestion & Orchestration Layer (ADF + ADLS), Presentation Layer (Azure SQL DB), and Business Layer (Power BI).

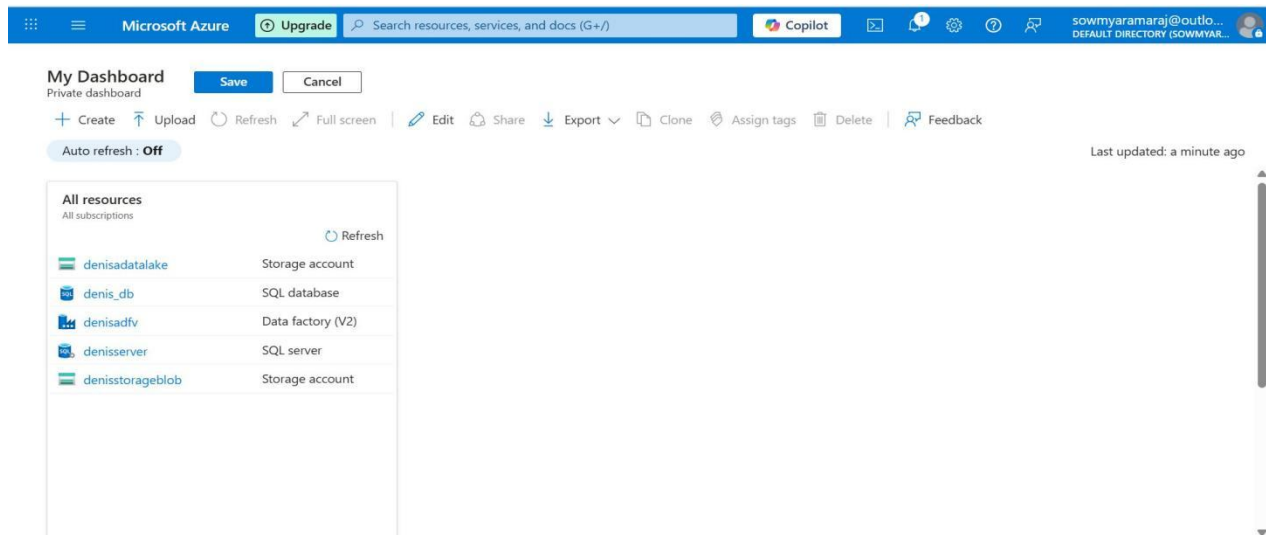


Figure 1: Azure Portal Dashboard — All Resources in the Project

The five core Azure resources provisioned for this project are:

- denisadatalake — Azure Data Lake Storage Gen2 (Medallion layers)
- denisstorageblob — Azure Blob Storage (raw source files)
- denisadfv — Azure Data Factory V2 (orchestration engine)
- denisserver — Azure SQL Server (logical server)
- denis_db — Azure SQL Database (gold-layer analytical target)

4. Data Sources

Data for this project is sourced from four distinct systems, each representing a different integration pattern:

Source System	Format	Data Entity	Location
Azure Data Lake Storage Gen2	CSV (by year)	Sales Transactions (2014–2017)	denisfactraw/sales/
Azure Blob Storage	CSV / XLSX	Categories, SubCategories, SalesRep, Geography	denisstorageblob/denis-raw-data/
MySQL (On-Premises)	Relational Table	Product	Local MySQL server via Self-Hosted IR

4.1 Azure Blob Storage — Raw Source Files

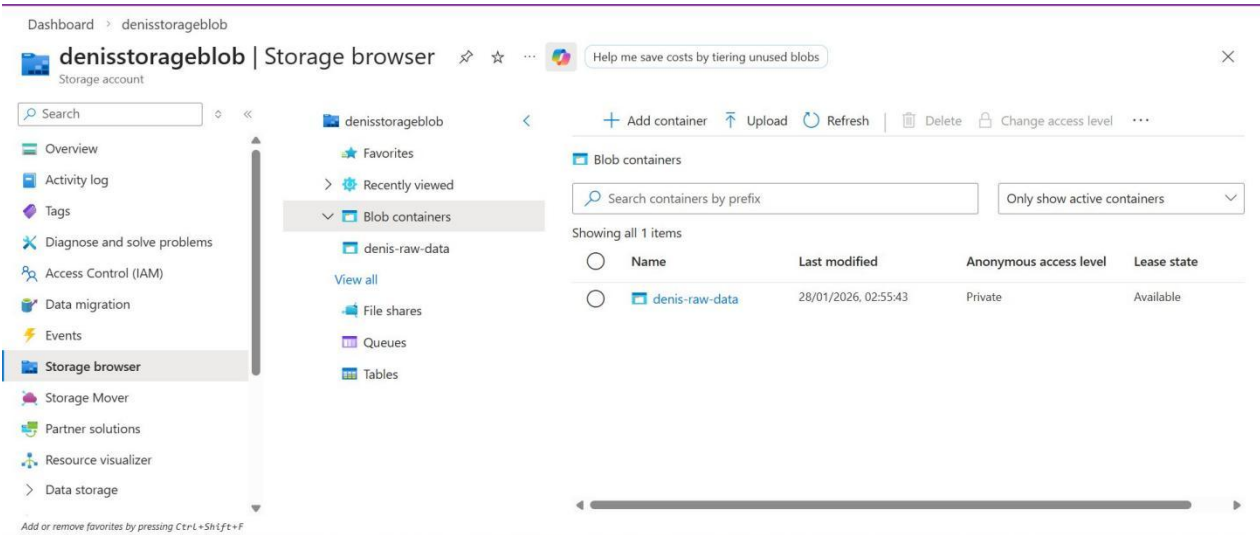


Figure 2: Blob Storage (denisstorageblob) — Raw Input Container 'denis-raw-data'

The Blob Storage account 'denisstorageblob' contains a single container 'denis-raw-data' holding four CSV files: Categories.csv, Geography.csv, SalesRep.csv, and SubCategories.csv. These are dimension-type files uploaded as raw inputs to the pipeline.

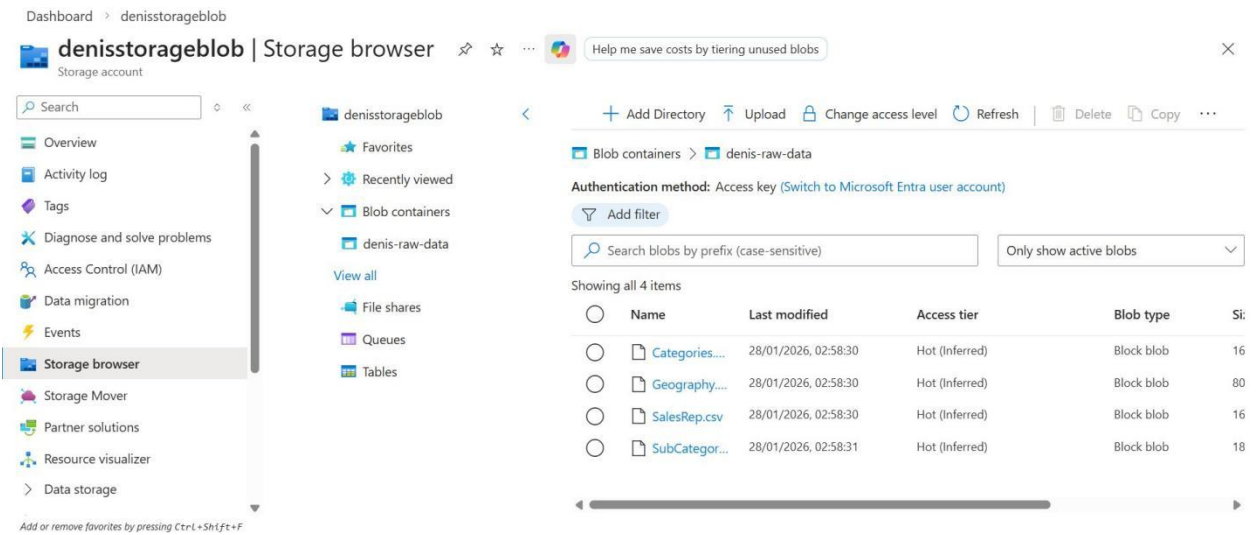


Figure 3: Raw Data Container Contents — Categories, Geography, SalesRep, SubCategories

4.2 Azure Data Lake Storage Gen2 — Sales Files

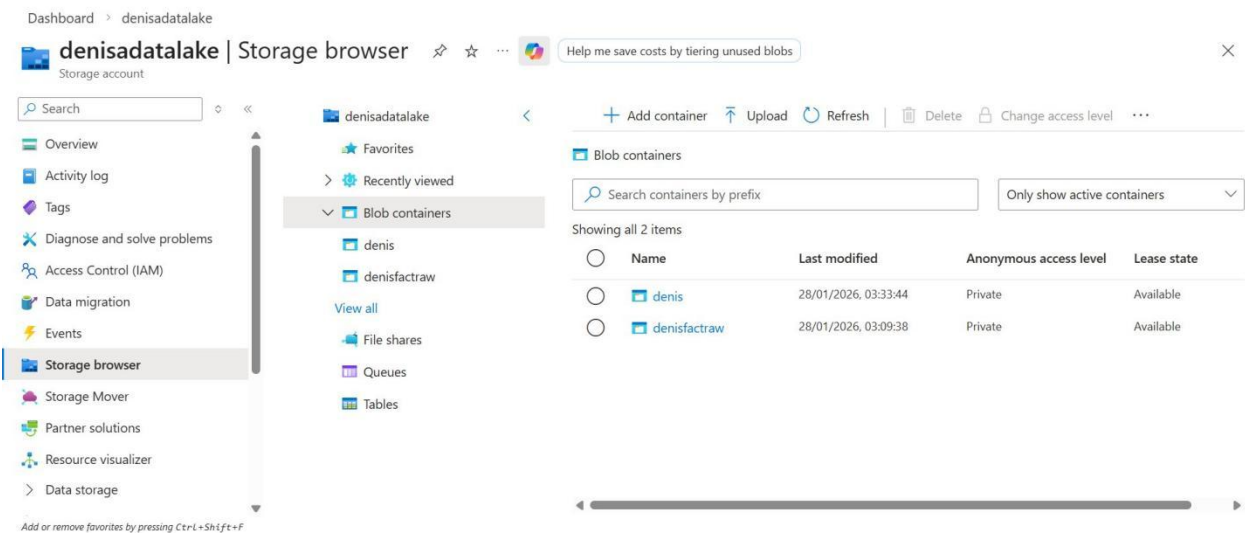


Figure 4: ADLS Gen2 (denisadatalake) — Blob Containers: 'denis' and 'denisfactraw'

The ADLS Gen2 account 'denisadatalake' holds two primary containers: 'denis' (the Medallion layers) and 'denisfactraw' (raw sales CSV files partitioned by year). The hierarchical namespace is enabled, providing full directory-level access control.

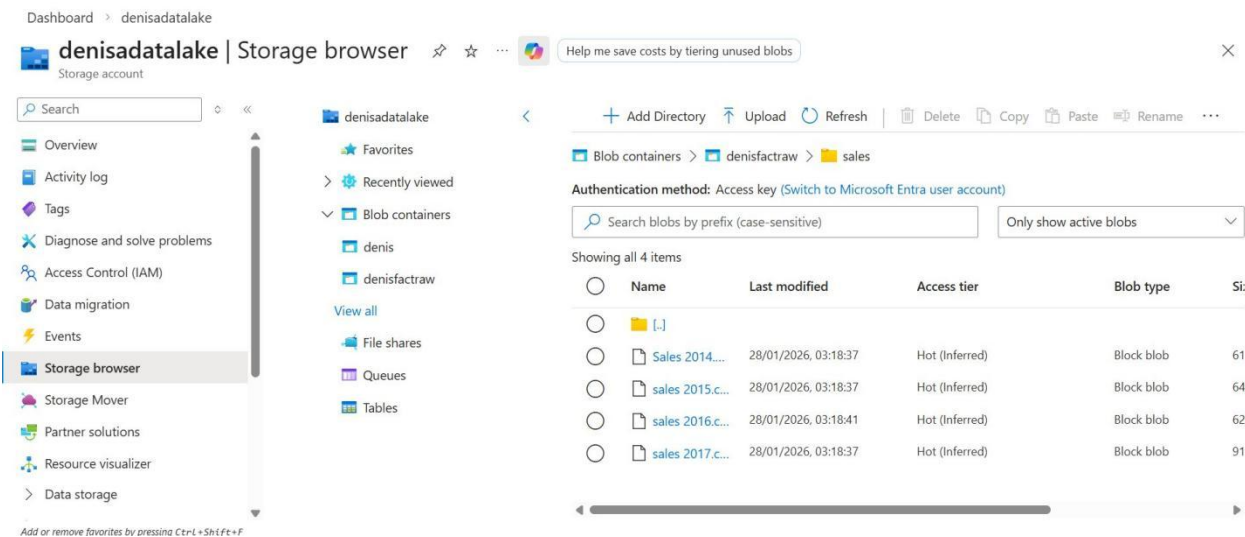


Figure 5: Raw Sales Data — denisfactraw/sales/ — Four yearly CSV files (2014–2017)

5. Azure Resources Setup

5.1 SQL Server & SQL Database

An Azure SQL Server ('denisserver') and SQL Database ('denis_db') were provisioned in the West US 2 region under resource group 'denis-rg'. The database operates on the Basic pricing tier and is used as the downstream target for Gold-layer data.

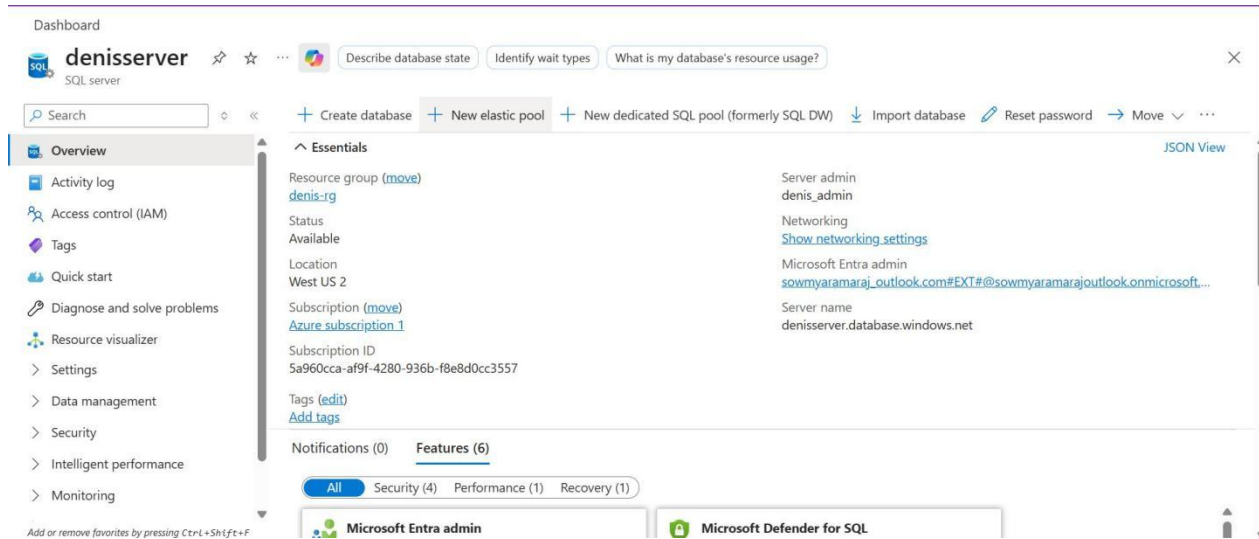


Figure 6: Azure SQL Server (denisserver) — Overview with admin, networking and Entra admin settings

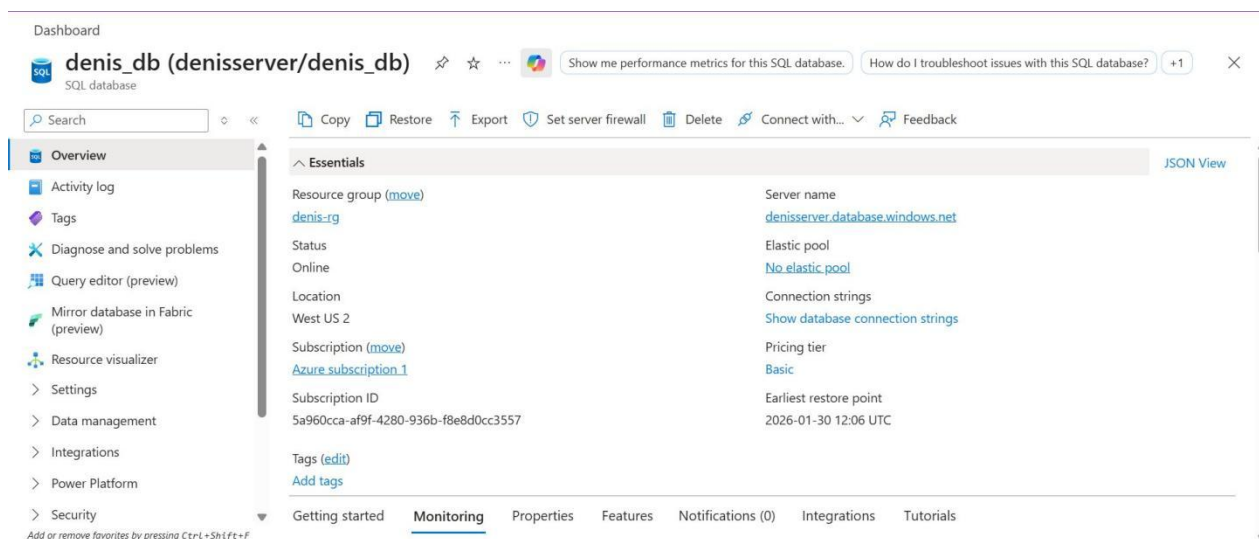


Figure 7: Azure SQL Database (denis_db) — Essentials: Status Online, Basic Tier, West US 2

5.2 Medallion Container Structure

The ADLS container 'denis' contains a folder named 'medallion' with three sub-folders corresponding to the three processing tiers of the Medallion Architecture.

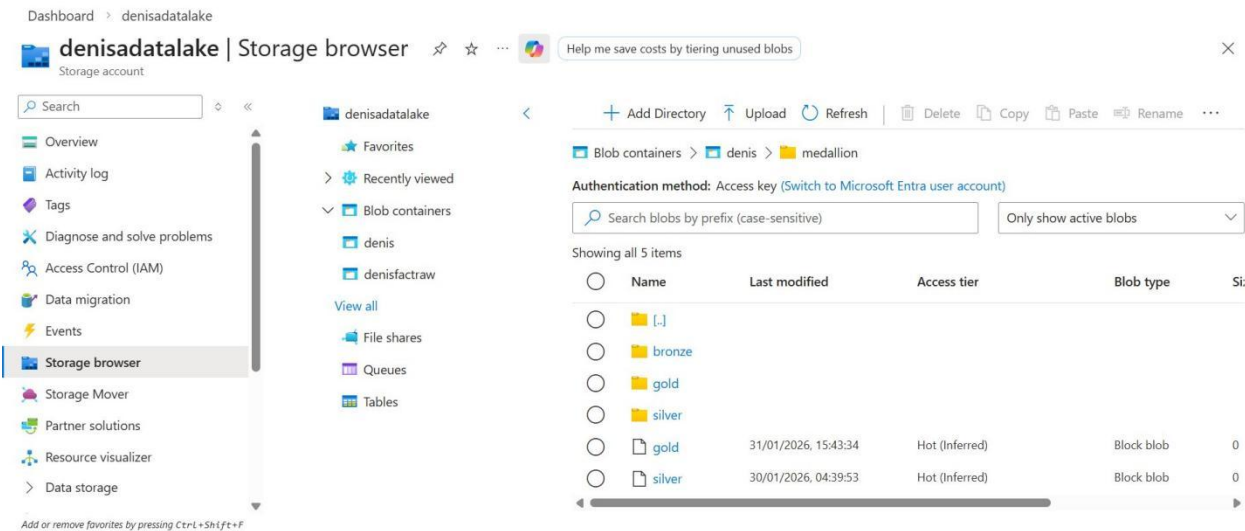


Figure 8: Medallion Container Structure — Bronze, Silver, Gold folders inside 'denis/medallion'

6. Azure Data Factory Configuration

6.1 Integration Runtimes

Azure Data Factory requires an Integration Runtime (IR) to serve as the compute infrastructure for data movement and transformation. Two IRs are configured in this project:

Runtime Name	Type	Sub-Type	Status	Purpose
AutoResolveIntegrationRuntime	Azure	Public	Running	Default cloud IR for Azure-to-Azure pipelines
selfhostedir	Self-Hosted	—	Running	Connects to on-premises MySQL database

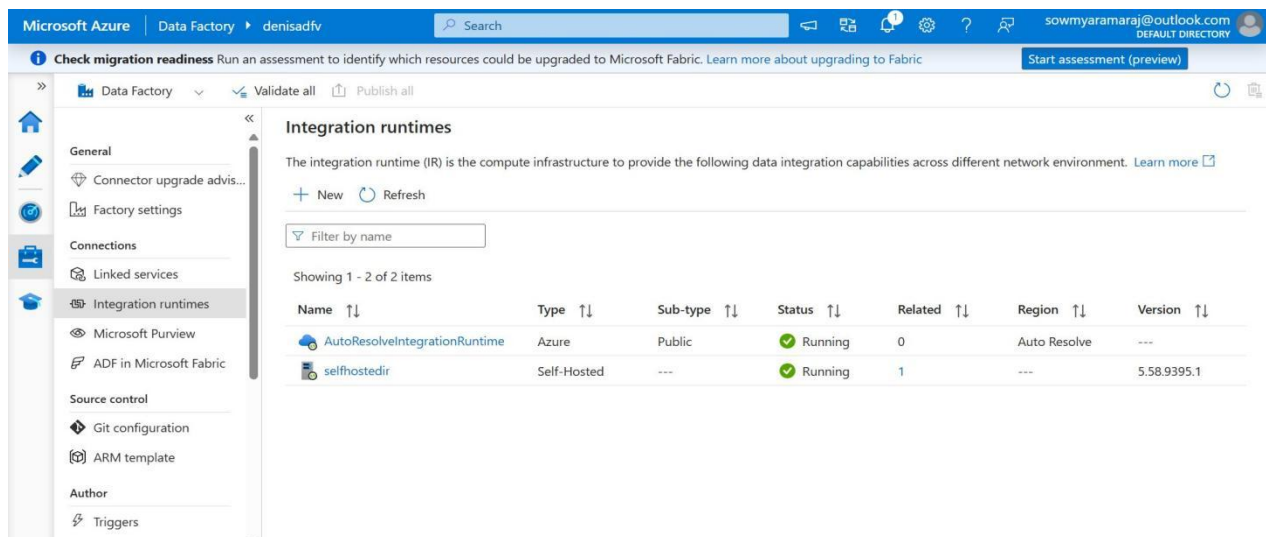


Figure 9: Integration Runtimes — AutoResolve (Azure) and selfhostedir (Self-Hosted, Running)

The Self-Hosted IR 'selfhostedir' (version 5.58.9395.1) is installed on the local machine running the MySQL instance. This enables ADF to securely reach on-premises data without requiring public internet exposure.

6.2 Linked Services

Linked Services define the connection strings to external data stores. Four linked services are configured, each serving a distinct integration endpoint:

Linked Service Name	Type	Purpose
adf2asqldb_ls	Azure SQL Database	Read/write to Azure SQL DB (denis_db)
adls2adf_ls	Azure Data Lake Storage Gen2	Access ADLS Medallion layers (Bronze/Silver/Gold)
blob2adf_ls	Azure Blob Storage	Read raw files from Blob container (denis-raw-data)
onprem2adls_ls	MySQL	Connect to on-premises MySQL via Self-Hosted IR

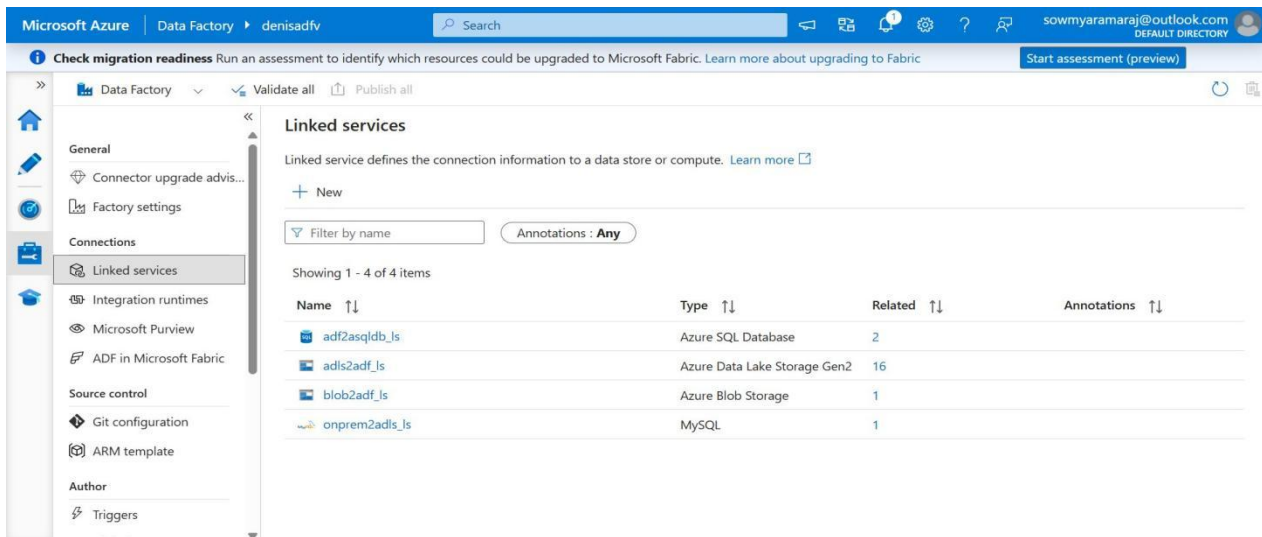


Figure 10: Linked Services — Four connections to SQL DB, ADLS Gen2, Blob Storage, and MySQL

7. Pipelines, Datasets & Data Flows

7.1 Factory Resources Overview

The ADF instance 'denisadf' contains pipelines, datasets, and data flows organized to implement the three-layer Medallion transformation. The following screenshots show the complete Factory Resources panel.

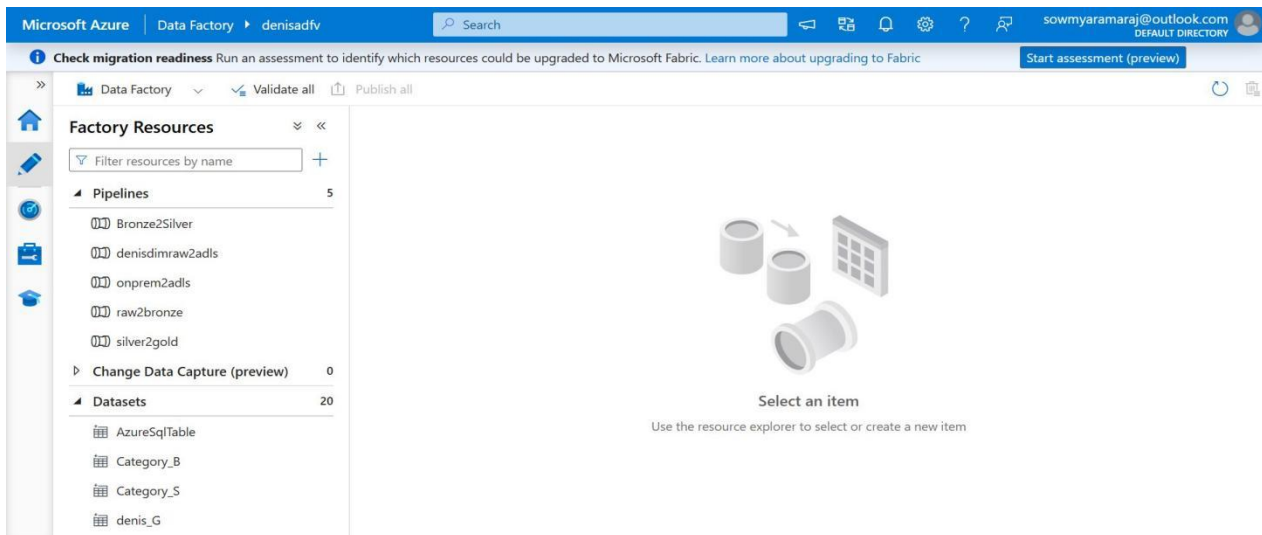


Figure 11: ADF Factory Resources — 5 Pipelines and first set of Datasets (AzureSqlTable, Category_B, Category_S, denis_G)

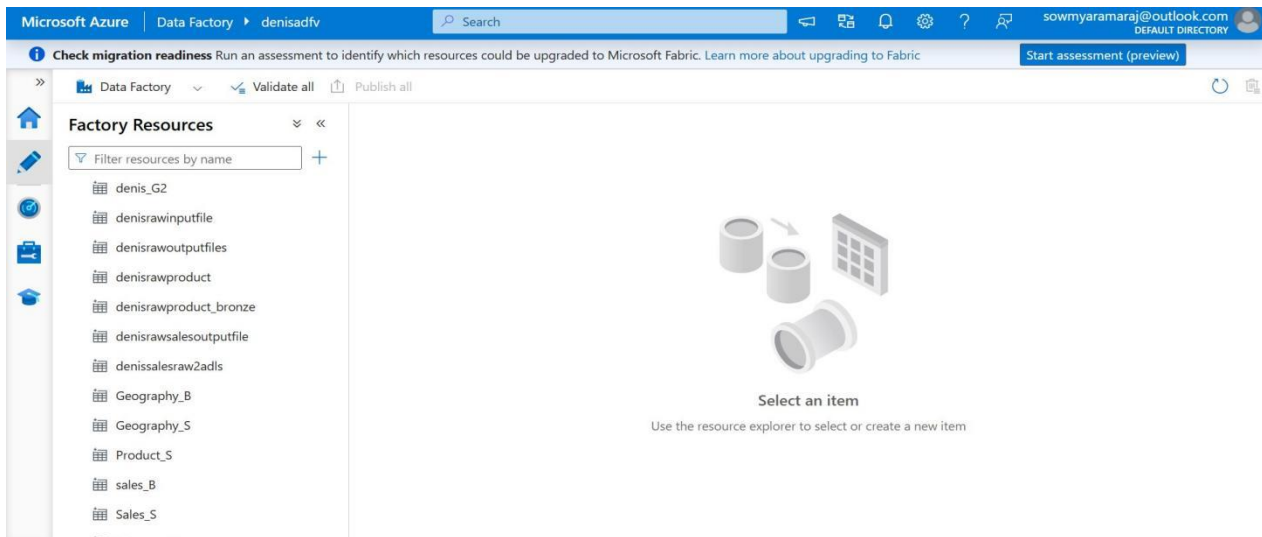


Figure 12: ADF Factory Resources — Continued Dataset list (denis_G2, denisrawinputfile, denisrawoutputfiles, denisrawproduct, denisrawproduct_bronze, denisrawsalesoutputfile, denissalesraw2adls, Geography_B, Geography_S, Product_S, sales_B, Sales_S)

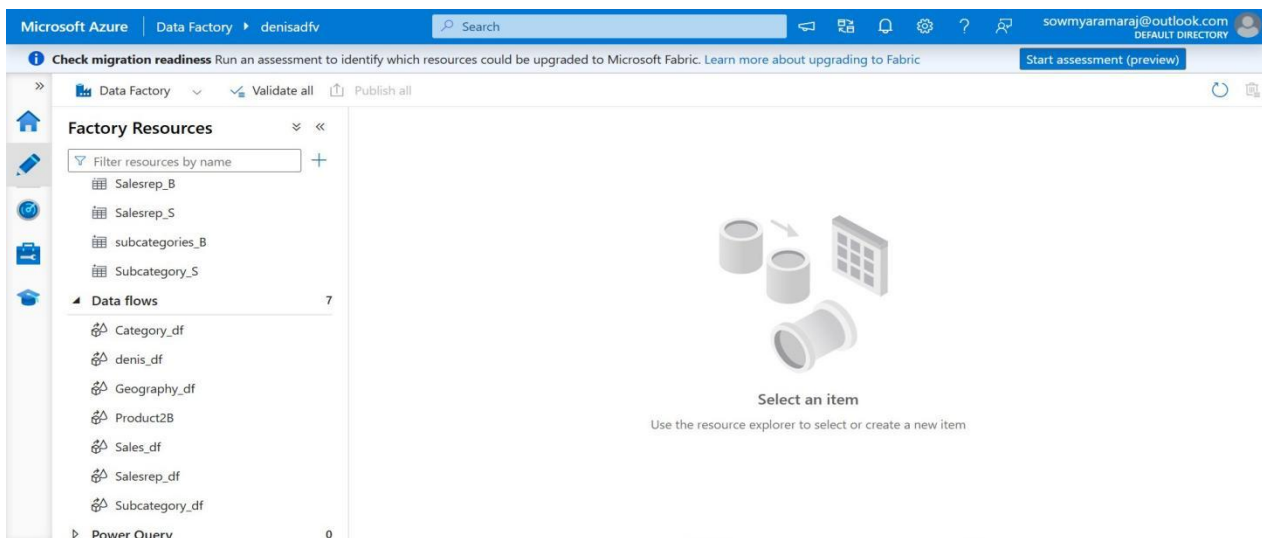


Figure 13: ADF Factory Resources — Final Datasets (Salesrep_B, Salesrep_S, subcategories_B, Subcategory_S) and 7 Data Flows (Category_df, denis_df, Geography_df, Product2B, Sales_df, Salesrep_df, Subcategory_df)

7.2 Pipeline Summary

Pipeline Name	Layer	Description
raw2bronze	Raw → Bronze	Copies files from Blob Storage (Categories, SubCategories, SalesRep) and ADLS (Sales CSVs) into the Bronze layer with standardized naming
onprem2adls	Raw → Bronze	Ingests Product data from on-premises MySQL via Self-Hosted IR into ADLS Bronze
Bronze2Silver	Bronze → Silver	Executes data flows to cleanse and transform all dimension and fact files into Silver
silver2gold	Silver → Gold	Runs the Denis_DF data flow to join all Silver entities and produce a single Gold file (Denis_G.csv) with business KPIs
denisdimraw2adls	Orchestration	Master pipeline coordinating the full end-to-end flow with dependency chaining

8. Medallion Architecture — Layer Walkthrough

8.1 Bronze Layer — Raw Ingestion

The Bronze layer receives data copied directly from source systems with minimal transformation. The goal is faithful landing of source data with standardized file naming. The pipeline uses wildcard copy patterns to pick up all files from the source containers.

Fixes applied during Raw-to-Bronze ingestion:

- Sales CSV files (Sales2014.csv through Sales2017.csv) are routed to a dedicated 'sales/' sub-folder within Bronze
- The MySQL Product table is written as 'Product.csv'

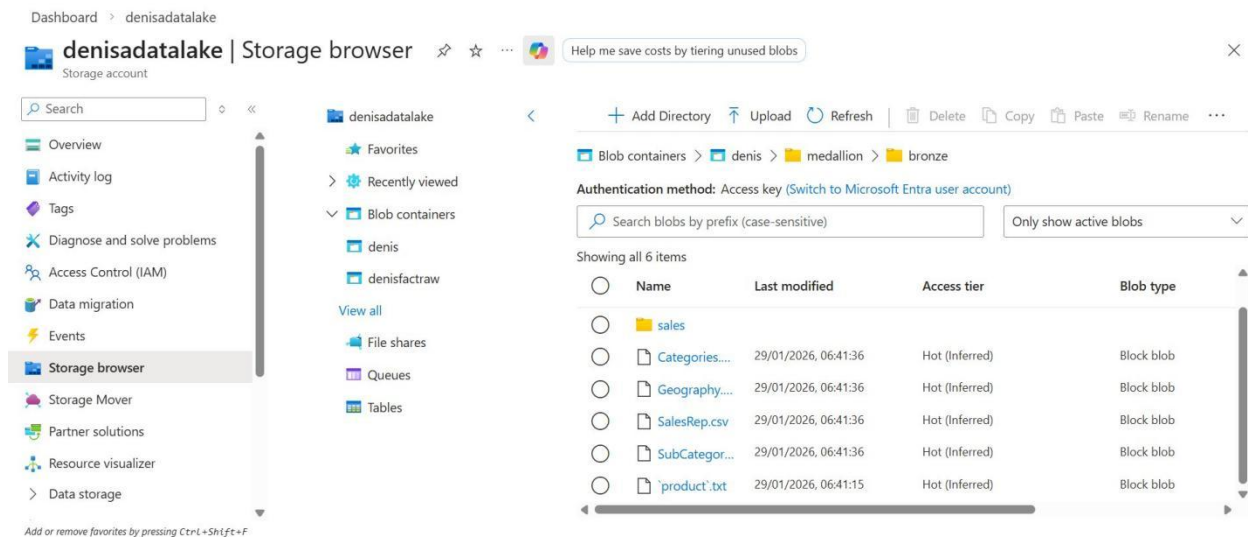


Figure 14: Bronze Layer Contents — sales/ folder, Categories.csv, Geography.csv, SalesRep.csv, SubCategories.csv, product.txt (6 items)

8.2 Silver Layer — Cleansed & Conformed

The Silver layer applies data quality transformations using ADF Mapping Data Flows. Each source entity has a dedicated data flow with the following transformation logic:

Entity	Transformation Applied	Output File
Category	Select required columns, standardize header names	Category_s.csv
SubCategory	Remove 'ID-' prefix from SubCategoryKey using substring(SubCategoryKey,5,9999). Inner join with Category on CategoryKey	Subcategory_s.csv
Geography	Add surrogate key (GeoKey), remove Wikipedia column, reorder columns: GeoKey, Country, Town	Geography_s.csv
Product	Import projection, deduplicate using Aggregate + Group By on all columns (duplicate records removed using aggregation logic), remove count column via Select	Product_s.csv
SalesRep	Read XLSX (Sales rep sheet), remove ID prefix from SalesRepKey	Salesrep_s.csv
Sales	Read all yearly CSVs, split Location column into Country and State using split(Location,','), convert Date using toDate(Date,'dd.MM.yyyy')	Sales_S.csv

All Silver sinks use 'Single Partition' in Optimize settings and 'Pattern' for File Name Option to produce a single, predictably named output file per entity.

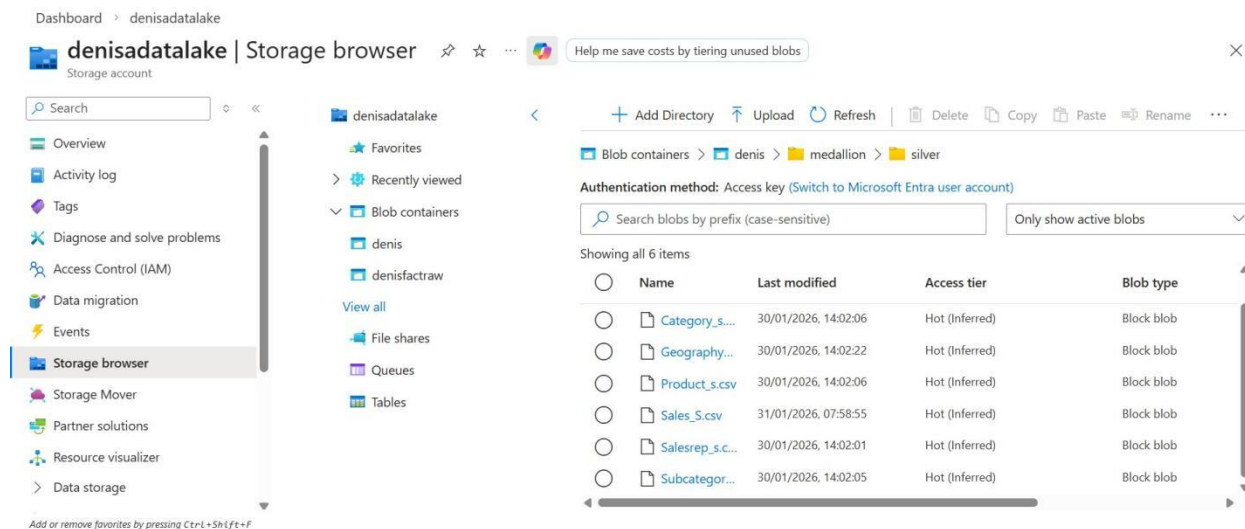


Figure 15: Silver Layer Contents — Category_s, Geography_s, Product_s, Sales_S, Salesrep_s, Subcategory_s (6 cleansed files)

8.3 Gold Layer — Business-Ready Analytics

The Gold layer produces a single denormalized analytical file (Denis_G.csv) by joining all Silver entities into one comprehensive dataset. The Denis_DF data flow performs the following join sequence:

1. Join Category + SubCategory on CategoryKey → catsubcat (Inner Join)
2. Join Product + catsubcat on SubCategoryKey → productcatsubcat (Inner Join)
3. Join Sales + SalesRep on SalesRepID → salessalesrepselect (Inner Join)
4. Join productcatsubcat + Geo on GeoKey → final product+geo join (Inner Join)
5. Final join: product-geo + salessalesrep → complete 15-column dataset

Calculated KPI columns added in the derived column transformation:

- TotalRevenue = RetailPrice * Units
- TotalCost = StandardCost * Units
- GrossProfit = TotalRevenue - TotalCost

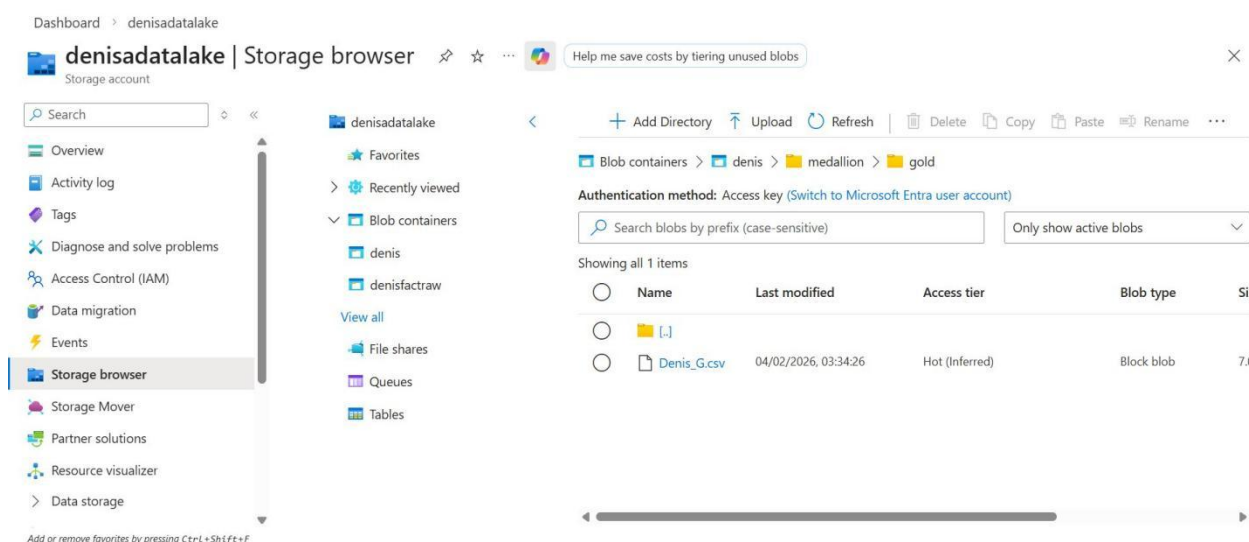


Figure 16: Gold Layer Contents — Denis_G.csv (single denormalized analytical file, 7.0 KB)

9. Data Modeling Approach

The data model follows a Star Schema design, with one central fact table surrounded by dimension tables. This design optimizes query performance for analytical workloads in Power BI and Azure SQL Database.

10. Conclusion

This project successfully demonstrates a production-grade, end-to-end Azure Data Engineering pipeline following modern best practices. The solution ingests heterogeneous sales data from distinct source systems, applies progressive data quality improvements across the Medallion Architecture layers, and delivers a business-ready analytical dataset enriched with revenue KPIs.

Key engineering strengths demonstrated by this implementation:

- **Scalability:** The wildcard-based ingestion patterns and parameterizable datasets allow new yearly sales files to be processed without pipeline modification.
- **Data Quality:** Each Medallion layer enforces specific quality rules — from raw landing in Bronze, through cleaning and deduplication in Silver, to denormalization and KPI enrichment in Gold.
- **Hybrid Connectivity:** The Self-Hosted Integration Runtime enables seamless connectivity to the on-premises MySQL source without compromising network security.
- **Observability:** ADF's built-in monitoring provides activity-level run status, duration, and integration runtime details for each pipeline execution.

- Business Value: The final Denis_G.csv dataset, combined with Azure SQL Database as the serving layer, directly enables Power BI dashboards for sales analysis by product, geography, category, and sales representative.

This implementation serves as a reusable reference architecture for Azure-based data warehousing projects involving multi-source ingestion, Medallion layer processing, and dimensional modelling for BI consumption.